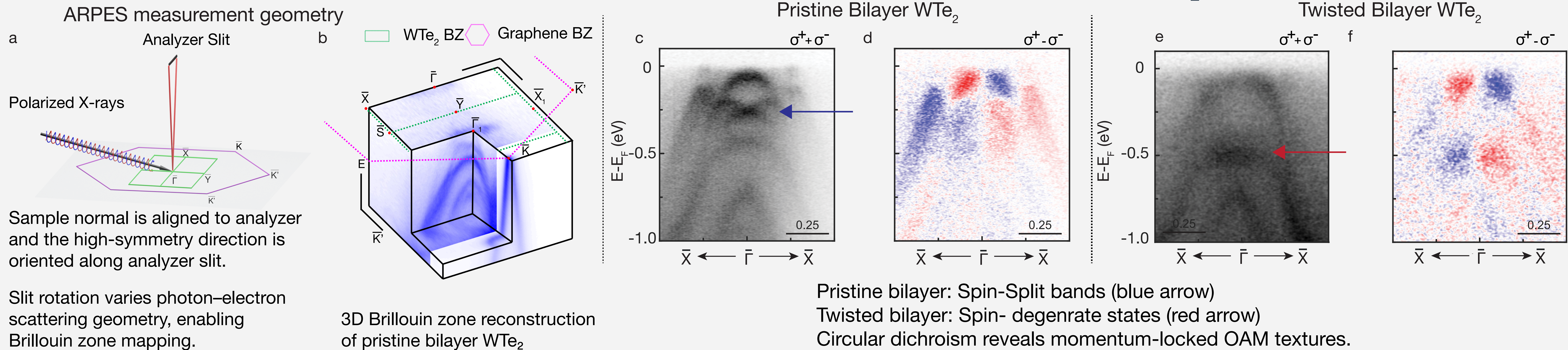
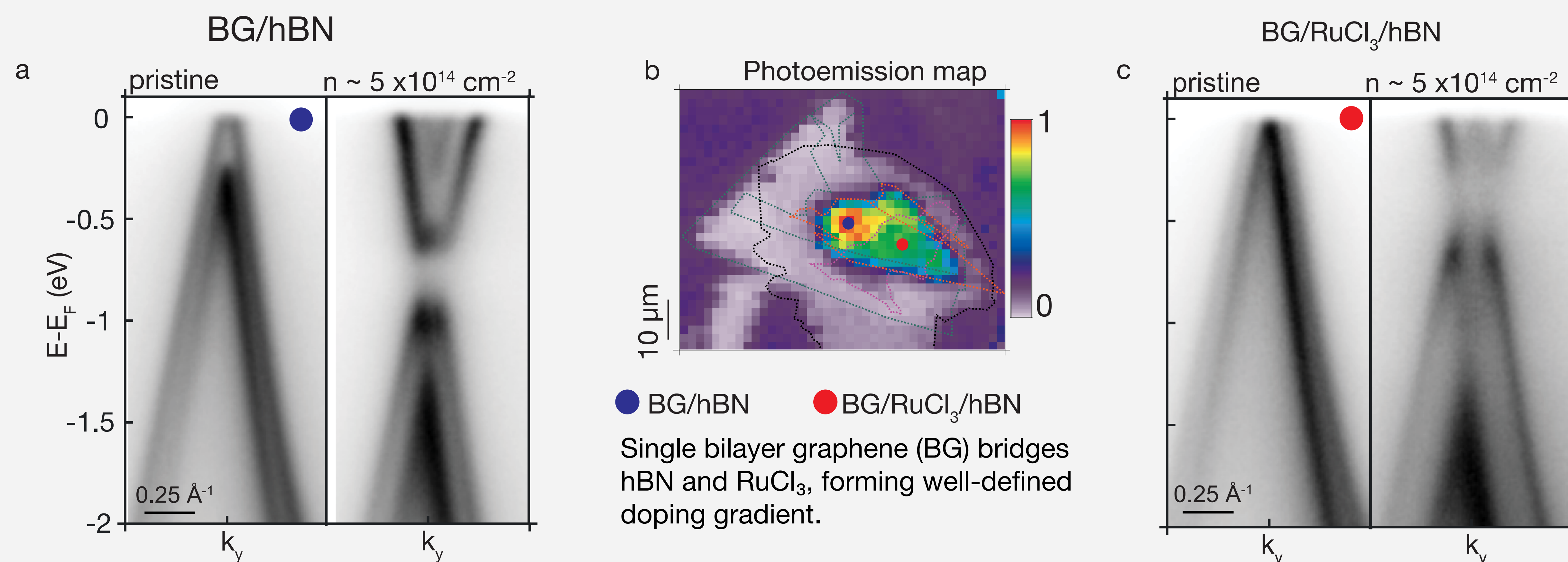


Abstract : Two-dimensional (2D) superlattices host a rich landscape of tunable correlated and topological quantum states. Angle-resolved photoemission spectroscopy with nanometer spatial resolution (nanoARPES) provides direct, momentum-resolved access to their electronic wavefunctions. Here, we investigate how twist angle, substrate coupling, out-of-plane electric fields, and light polarization can modify the electronic structure of 2D superlattices.

ARPES Circular Dichroism in Pristine and Aligned Bilayer WTe₂

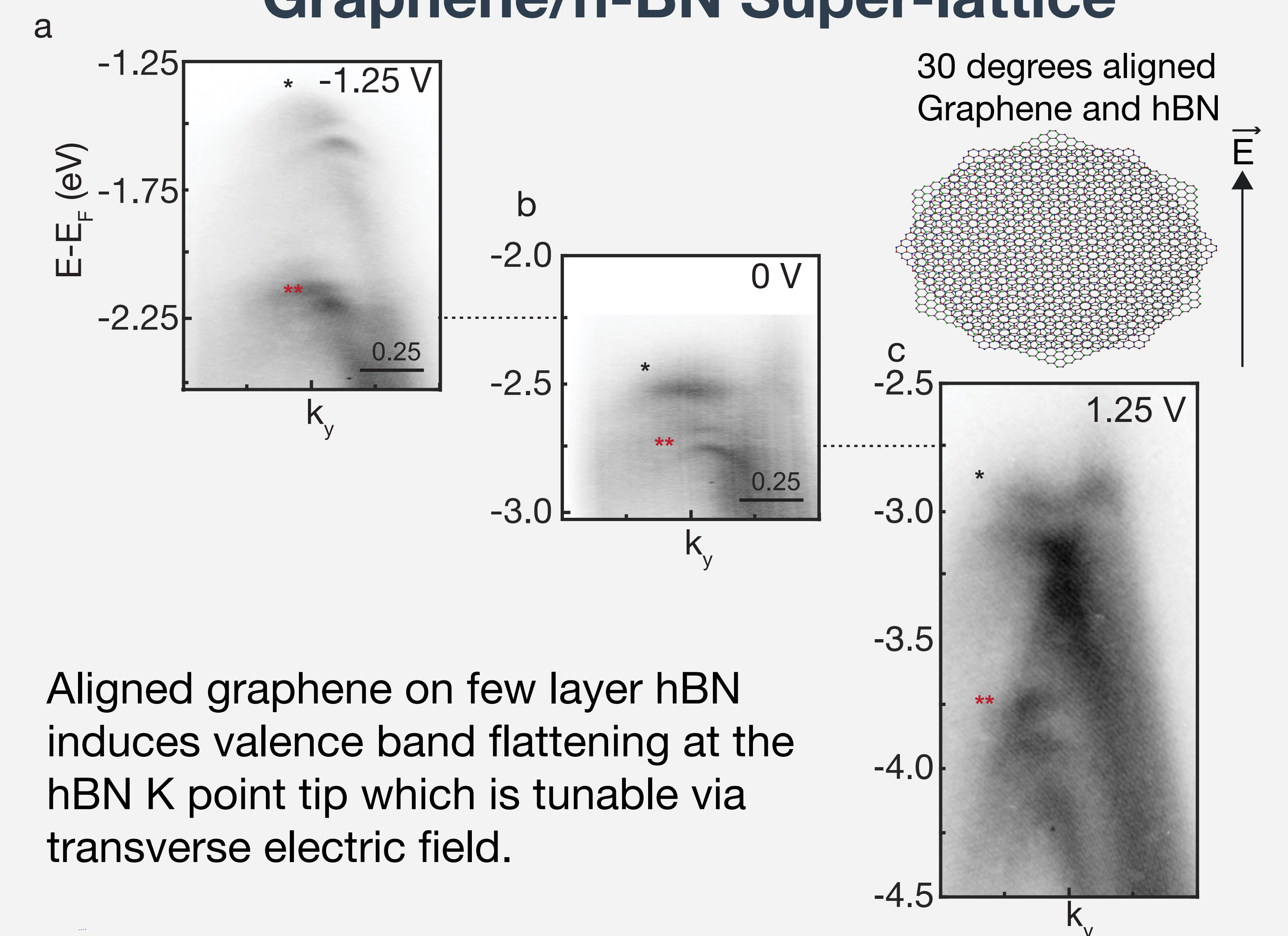


Substrate-Driven Electronic Reconstruction in Bilayer Graphene



Additional carrier dosing opens a gap at the Dirac point in BG/hBN
Substrate-induced strain stabilizes mixed AA+AB stacking in BG/RuCl₃

Electric Field tunable Correlated States in Graphene/h-BN Super-lattice



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Acknowledgements: We acknowledge funding from the U.S. Department of Energy, Office of Basic Sciences, under Awards DE-SC0020323 and DE-SC0025490, and from the National Science Foundation (NSF) through Award No. DMR2339309 and the NSF Early Career Award No. 2339309.